

paper yoyos quasi-static model

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August 18, 2026

1 Geometric model

The system can be modelled as a constant-width rectangular ribbon of negligible thickness wrapped around a uniform-radius cylinder at a uniform pitch. One end of the ribbon is fixed in place to prevent rigid body motion, while the other (“active” end) is displaced in the z direction (the axis of the cylinder). Refer to Figure 1 for a visual representation of the system. The system has the following geometric constants:

- L : the length of the unrolled rectangle.
- W : the width of the unrolled rectangle. Assume $L \gg W$.
- θ_L : the total wind angle of the helix, which is fixed based on the boundary conditions (the two ends of the ribbon cannot rotate about the z -axis relative to one another, only translate along it, so the total number of turns is a fixed topological property of how the ribbon was wound and does not change as it is pulled). $\theta_L = 2\pi n$ where n is the number of winds around the z axis (not necessarily integer).

The state of the system is parametrized by the parameter x , which describes how far the active end has displaced in the z direction relative to the base end.

1.1 Coordinate systems

Positions on the helix can be described with local surface u, v coordinates, which describe the unrolled rectangle. The u direction runs along the length of the unrolled rectangle, and the v direction runs across its width. Alternatively, positions can be described in 3d space using cylindrical r, θ, z coordinates.

To simplify the geometry, we assume that the helix has a constant radius r and a constant pitch angle ϕ (the angle the local $\hat{u}$ makes with the r, θ plane). Equivalently, the local orthonormal frame $(\hat{u}, \hat{v})$ tangent to the ribbon surface is related to the cylinder’s own natural frame $(\hat{\theta}, \hat{z})$ by a rotation through the angle ϕ :

$$\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}, \quad \hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}. \quad (1)$$

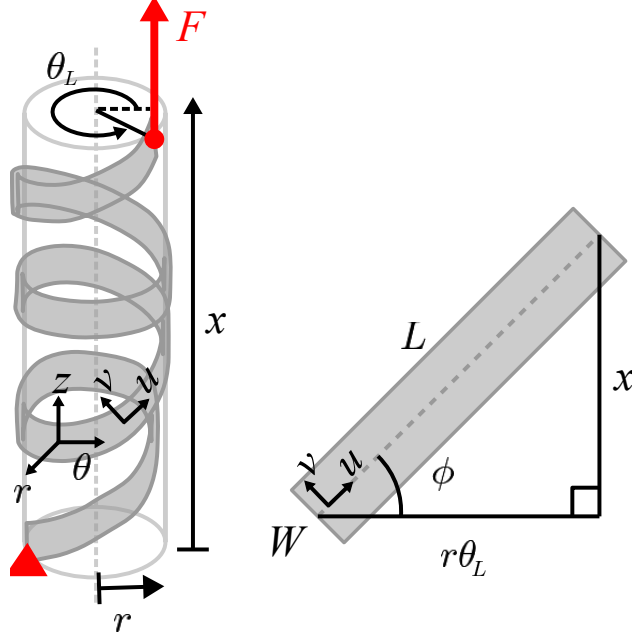


Figure 1: in progress

We now use Eq. (1) to relate the geometric constants L, θ_L to the pitch angle ϕ and radius r at a given extension x . From Eq. (1), $\hat{u}$ has z -component $\sin \phi$, so moving a distance du along the ribbon's own length produces an axial displacement $dz = \sin \phi du$. Since ϕ is constant along the whole ribbon at a given (frozen) extension x , integrating this from the base end ($u = 0$) to the active end ($u = L$) gives a total axial displacement of $L \sin \phi$. By definition, this total axial displacement between the two ends of the ribbon is exactly x , so

$$x = L \sin \phi \quad \implies \quad \phi = \sin^{-1} \left(\frac{x}{L} \right). \quad (2)$$

Likewise, $\hat{u}$ has θ -component $\cos \phi$, so moving a distance du along the ribbon produces a circumferential arc length $r d\theta = \cos \phi du$. Integrating over the whole ribbon length L , the ribbon sweeps out a total circumferential arc length $L \cos \phi$. This total arc length must also equal $r \theta_L$ (arc length equals radius times swept angle), since by construction the ribbon winds through the fixed total angle θ_L regardless of x . Equating the two expressions for this arc length and using $\cos \phi = \sqrt{1 - \sin^2 \phi}$ together with Eq. (2) gives

$$r \theta_L = L \cos \phi \quad \implies \quad r = \frac{L}{\theta_L} \cos \phi = \frac{L}{\theta_L} \sqrt{1 - \frac{x^2}{L^2}} = \frac{\sqrt{L^2 - x^2}}{\theta_L}. \quad (3)$$

1.2 Position vector

An area element $du \times dv$ at point (u, v) has a position vector given in cylindrical coordinates. We build this up from Eq. (1): since $\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}$, moving along the ribbon by du produces $r d\theta = \cos \phi du$ and $dz = \sin \phi du$; since $\hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}$, moving across the width by dv produces $r d\theta = -\sin \phi dv$ and $dz = \cos \phi dv$. Integrating these (at fixed, frozen x , so r and ϕ are

constants) from the reference point $(u, v) = (0, 0)$ gives

$$\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u \cos \phi - v \sin \phi}{r} \\ u \sin \phi + v \cos \phi \end{bmatrix}. \quad (4)$$

1.3 Local curvature

A cylinder of radius r has two principal curvatures: $\kappa_\theta = 1/r$ in the circumferential ($\hat{\theta}$) direction in which it is rolled, and $\kappa_z = 0$ in the axial ($\hat{z}$) direction in which it is completely flat. Equivalently, the standard cylindrical basis vectors rotate with θ at the rates

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}, \quad \frac{\partial \hat{z}}{\partial \theta} = 0, \quad (5)$$

(the first two follow from writing $\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ and $\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$ in terms of the fixed Cartesian basis and differentiating; the third holds because $\hat{z}$ does not depend on θ at all — this is exactly the statement $\kappa_z = 0$).

To differentiate the ribbon's own frame $(\hat{u}, \hat{v}, \hat{r})$ with respect to (u, v) we use the chain rule through θ . From $\theta(u, v) = (u \cos \phi - v \sin \phi)/r$ (Eq. 4), holding x — hence ϕ, r — fixed,

$$\frac{\partial \theta}{\partial u} = \frac{\cos \phi}{r}, \quad \frac{\partial \theta}{\partial v} = -\frac{\sin \phi}{r}. \quad (6)$$

Combining Eq. (5) and Eq. (6) via the chain rule (e.g. $\partial \hat{r} / \partial u = (\partial \hat{r} / \partial \theta)(\partial \theta / \partial u)$) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{r}}{\partial v} = -\frac{\sin \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos \phi}{r} \hat{r}, \quad \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi}{r} \hat{r}, \quad (7)$$

with $\hat{z}$ constant in u, v throughout. Now differentiate $\hat{u}, \hat{v}$ themselves, using $\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}$, $\hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}$ (Eq. 1) with ϕ held fixed:

$$\begin{aligned} \frac{\partial \hat{u}}{\partial u} &= \cos \phi \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos^2 \phi}{r} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \cos \phi \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi \cos \phi}{r} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial u} = \frac{\sin \phi \cos \phi}{r} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial v} = -\frac{\sin^2 \phi}{r} \hat{r}. \end{aligned} \quad (8)$$

The same three coefficients, $\cos^2 \phi / r$, $\sin^2 \phi / r$, and $\sin \phi \cos \phi / r$ (the last one twice), keep reappearing; we name them for convenience,

$$\kappa_{uu} := \frac{\cos^2 \phi}{r}, \quad \kappa_{vv} := \frac{\sin^2 \phi}{r}, \quad \kappa_{uv} := \frac{\sin \phi \cos \phi}{r}, \quad (9)$$

so that Eq. (8) reads $\partial \hat{u} / \partial u = -\kappa_{uu} \hat{r}$, $\partial \hat{u} / \partial v = \kappa_{uv} \hat{r}$, $\partial \hat{v} / \partial u = \kappa_{uv} \hat{r}$, $\partial \hat{v} / \partial v = -\kappa_{vv} \hat{r}$.

What κ_{uv} means. κ_{uu} and κ_{vv} are the ordinary curvatures of the u - and v -coordinate curves themselves: κ_{uu} is how fast $\hat{u}$ tips toward $\hat{r}$ per unit distance moved *along* $\hat{u}$, and likewise κ_{vv} for

$\hat{v}$ along $\hat{v}$. κ_{uv} is instead a *cross* rate: how fast $\hat{u}$ tips toward $\hat{r}$ per unit distance moved in the v -direction — equivalently, how fast $\hat{v}$ tips toward $\hat{r}$ per unit distance moved in the u -direction, since Eq. (8) gives the same coefficient for both. It is not a slope like dv/du ; it has the same units (one over length) and the same geometric meaning as κ_{uu}, κ_{vv} — a rate at which a tangent direction tilts toward the surface normal — just for a mixed pair of directions rather than a single one.

Concretely, κ_{uv} arises above because $\hat{u}, \hat{v}$ are each a mix of $\hat{\theta}$ and $\hat{z}$ (Eq. 1), and only the $\hat{\theta}$ part of either vector rotates as θ changes ($\hat{z}$ does not, Eq. 5). Moving in v changes θ (Eq. 6), and therefore rotates the $\hat{\theta}$ -component of $\hat{u}$ toward $\hat{r}$ even though v is not $\hat{u}$'s own direction — that leftover rotation is exactly κ_{uv} . Put differently: $\hat{u}, \hat{v}$ are rotated by ϕ away from the cylinder's own principal directions $\hat{\theta}, \hat{z}$, which have no such cross term ($\hat{\theta}$ only ever rotates toward $\hat{r}$ as θ itself changes, never as z changes, and $\hat{z}$ never rotates at all); κ_{uv} is exactly the off-diagonal curvature this rotation introduces. Consistent with this, $\kappa_{uv} = \sin \phi \cos \phi / r$ vanishes at $\phi = 0, 90^\circ$ (where $\hat{u}, \hat{v}$ realign with $\hat{\theta}, \hat{z}$) and is largest at $\phi = 45^\circ$.

Rotation of $\hat{r}$. It remains to differentiate $\hat{r}$ itself. We already have $\partial \hat{r} / \partial u, \partial \hat{r} / \partial v$ in terms of $\hat{\theta}$ (Eq. 7); to express them in terms of $\hat{u}, \hat{v}$ instead, invert Eq. (1) for $\hat{\theta}$ (solving its two equations for $\hat{\theta}, \hat{z}$ in terms of $\hat{u}, \hat{v}$ gives $\hat{\theta} = \cos \phi \hat{u} - \sin \phi \hat{v}$). Substituting,

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} \hat{\theta} = \frac{\cos \phi}{r} (\cos \phi \hat{u} - \sin \phi \hat{v}) = \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, \quad (10)$$

and the identical substitution into $\partial \hat{r} / \partial v = -\frac{\sin \phi}{r} \hat{\theta}$ gives $\partial \hat{r} / \partial v = -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}$. Collecting everything derived in this section,

$$\begin{aligned} \frac{\partial \hat{r}}{\partial u} &= \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, & \frac{\partial \hat{r}}{\partial v} &= -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}, \\ \frac{\partial \hat{u}}{\partial u} &= -\kappa_{uu} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \kappa_{uv} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= \kappa_{uv} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\kappa_{vv} \hat{r}. \end{aligned} \quad (11)$$

Two features of Eq. (11) are worth flagging because they are used repeatedly below: (i) $\hat{u}, \hat{v}$ never rotate into each other ($\partial \hat{u} / \partial u$ and $\partial \hat{u} / \partial v$ have no $\hat{v}$ -component, and vice versa) — this holds because the u - and v -coordinate lines are straight lines on the flat unrolled rectangle, hence remain geodesics after the isometric wrapping into a helix, and a curve's own tangent frame only rotates toward the surface normal along a geodesic, never within the tangent plane; (ii) the rotation matrix is antisymmetric, e.g. $\partial \hat{u} / \partial u \cdot \hat{r} = -\partial \hat{r} / \partial u \cdot \hat{u}$, as required for any rotating orthonormal frame.

2 Stress balance

We will model the paper yoyo as a thin shell of small but finite thickness $t \ll r$. Because the ribbon's thickness ultimately carries the boundary tractions responsible for friction (Section 3), we work throughout with its full three-dimensional stress state. We use $\zeta \in [-\frac{t}{2}, \frac{t}{2}]$ for the coordinate measuring position through the thickness, i.e., displacement from the mid-surface along $\hat{r}$: a general

material point is at $\vec{p}(u, v, x) + \zeta \hat{r}(u, v)$, with $\vec{p}(u, v, x)$ the mid-surface position of Eq. (4). At a point (u, v, ζ) the material carries the full Cauchy stress tensor

$$\sigma = \begin{pmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{pmatrix}. \quad (12)$$

There are six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears σ_{ur}, σ_{vr} , by which the ribbon can transmit a load sideways from one part of its width to another; and σ_{rr} , the normal pressure caused by layer-layer contact at the ribbon's surface.

We perform a force balance on a differential volume element $du \times dv \times d\zeta$. On each of its six faces there is a traction vector: on the u -face (outward normal $\hat{u}$), the traction is $\vec{t}_u = \sigma_{uu}\hat{u} + \sigma_{uv}\hat{v} + \sigma_{ur}\hat{r}$, and similarly $\vec{t}_v = \sigma_{uv}\hat{u} + \sigma_{vv}\hat{v} + \sigma_{vr}\hat{r}$ on the v -face and $\vec{t}_\zeta = \sigma_{ur}\hat{u} + \sigma_{vr}\hat{v} + \sigma_{rr}\hat{r}$ on the ζ -face. Because $t \ll r$, the element can be approximated as a rectangular prism, so the net force on the element is

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta} \right) du dv d\zeta = 0. \quad (13)$$

However, because $\hat{u}, \hat{v}, \hat{r}$ themselves depend on u, v (Eq. 11), differentiating $\vec{t}_u, \vec{t}_v, \vec{t}_\zeta$ produces extra terms beyond the plain derivatives of the stress components. For example, expanding $\frac{\partial}{\partial u}(\sigma_{uu}\hat{u})$ includes a curvature-weighted term in the $\hat{r}$ direction,

$$\frac{\partial}{\partial u}(\sigma_{uu}\hat{u}) = \frac{\partial \sigma_{uu}}{\partial u}\hat{u} + \sigma_{uu}\frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u}\hat{u} - \kappa_{uu}\sigma_{uu}\hat{r}, \quad (14)$$

the additional curvature-weighted terms reflecting the fact that the element is not perfectly planar, so in-plane stresses are slightly rotated to have a small out-of-plane component, and vice versa. Expanding every term this way (using Eq. 11 throughout) and sorting by component gives three scalar force-balance equations, exact to leading order in $\frac{t}{r}$:

$$\hat{u} : \boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu}\sigma_{ur} - \kappa_{uv}\sigma_{vr} = 0} \quad (15)$$

$$\hat{v} : \boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv}\sigma_{ur} + \kappa_{vv}\sigma_{vr} = 0} \quad (16)$$

$$\hat{r} : \boxed{\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu}\sigma_{uu} + 2\kappa_{uv}\sigma_{uv} - \kappa_{vv}\sigma_{vv} = 0} \quad (17)$$

Equilibrium alone gives three equations for six unknown stress components. Because t is small, however, we make a first-order approximation that each stress component varies linearly through t : knowing its two boundary values at $\zeta = \pm \frac{t}{2}$, for every (u, v) , is then equivalent to knowing the full through-thickness profile. Section 3 supplies exactly those boundary values for $\sigma_{ur}, \sigma_{vr}, \sigma_{rr}$, from layer-layer contact; Section 4 carries out the thickness integration itself, retaining — rather than dropping — the curvature-weighted terms above.

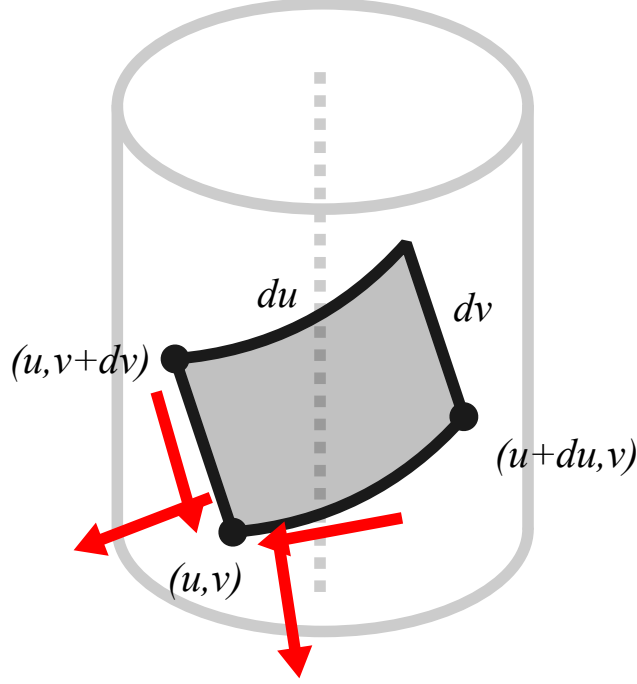


Figure 2: in progress

3 Friction

3.1 Overlapping neighbors

Because the ribbon is wound around itself, a given area element at (u, v) could physically overlap with another area element one full turn away along the ribbon. Specifically, an area element at point (u, v) overlaps with the neighboring interior layer at point (u^-, v^-) if their cylindrical coordinates have equal r and z , but differ by one full turn in θ . We can express this by taking the difference in $\vec{p}$:

$$\vec{p}(u, v, x) - \vec{p}(u^-, v^-, x) = \begin{bmatrix} 0 \\ 2\pi \\ 0 \end{bmatrix}. \quad (18)$$

Plugging in Eq. (4) gives

$$u^- = u - 2\pi r \cos \phi, \quad (19)$$

$$v^- = v + 2\pi r \sin \phi. \quad (20)$$

By the identical argument with $\theta(u^+, v^+) = \theta(u, v) + 2\pi$ (i.e., the neighboring *outer* layer, one full turn further out), the element has an outer overlapping neighbor at

$$u^+ = u + 2\pi r \cos \phi, \quad (21)$$

$$v^+ = v - 2\pi r \sin \phi. \quad (22)$$

We neglect edge effects where u^- or u^+ falls outside the $[0, L]$ bounds — there is simply no material wound that far, regardless of v . Within $[0, L]$, an inner neighbor is present exactly whenever v^- lies within $[-\frac{W}{2}, \frac{W}{2}]$, and an outer neighbor is present exactly whenever v^+ lies in $[-\frac{W}{2}, \frac{W}{2}]$, i.e.,

$$\text{Inner neighbor present} \iff v \leq \frac{W}{2} - 2\pi r \sin \phi, \quad (23)$$

$$\text{Outer neighbor present} \iff v \geq -\frac{W}{2} + 2\pi r \sin \phi. \quad (24)$$

3.2 Friction direction and magnitude

The position vector $\vec{p}(u, v, x)$ of a material point (u, v) is given by Eq. (4). Substituting Eq. (2) and Eq. (3) to expose the dependence on x gives

$$\vec{p}(u, v, x) = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u\theta_L}{L} - \frac{v x \theta_L}{L\sqrt{L^2 - x^2}} \\ \frac{ux}{L} + \frac{v\sqrt{L^2 - x^2}}{L} \end{bmatrix}. \quad (25)$$

The velocity of a given material point (u, v) as the active end of the ribbon is displaced (i.e., as x changes) is then obtained by taking the partial derivative of each component with respect to x :

$$\frac{\partial}{\partial x} \vec{p}(u, v, x) = \begin{bmatrix} \frac{\partial r}{\partial x} \\ \frac{\partial \theta}{\partial x} \\ \frac{\partial z}{\partial x} \end{bmatrix} = \begin{bmatrix} -\frac{x}{\theta_L \sqrt{L^2 - x^2}} \\ -\frac{L \theta_L v}{(L^2 - x^2)^{3/2}} \\ \frac{u}{L} - \frac{vx}{L\sqrt{L^2 - x^2}} \end{bmatrix}. \quad (26)$$

We can then plug u^-, v^- into Eq. (26) to compute the velocity of the element relative to its inner overlapping neighbor, which gives the relative slip direction in the r, θ, z frame:

$$\frac{\partial \vec{p}(u, v, x)}{\partial x} - \frac{\partial \vec{p}(u^-, v^-, x)}{\partial x} = \begin{bmatrix} 0 \\ \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix}. \quad (27)$$

As expected, there is no relative velocity in the radial direction (i.e., the layers are not lifting away from each other). We can now compute the relative slip direction in the local frame by projecting this slip velocity from the θ, z basis onto the u, v basis using the relation described by Eq. (1). Since $\partial\theta/\partial x$ above is an *angular* rate, we first convert it to a linear (arc-length) rate by multiplying by r , giving $v_\theta = r \partial\theta/\partial x$; then projecting as just described:

$$\vec{v}_{slip}(u, v, x) = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} r \cdot \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix}. \quad (28)$$

We simplify this expression by substituting x and r in terms of ϕ (Eq. (2), (3)) to expose the double-angle combination, then applying double-angle trig identities to factor out common terms.

The result is a scaled unit vector:

$$\vec{v}_{slip}(u, v, x) = \frac{2\pi}{\theta_L \cos \phi} \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix}. \quad (29)$$

The prefactor is strictly positive for all $x \in (0, L)$. Therefore, the relative slip velocity points in the direction of the unit vector $(\sin 2\phi, \cos 2\phi)$ in the local u, v frame, so the Coulomb friction force opposing this motion must point in the opposite direction:

$$\hat{f} = - \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix}. \quad (30)$$

Recall that this is the friction exerted on the element by the inner layer, which resists the element from sliding forward with x . The components of this friction force are equivalent to the shear stresses acting on the inner $(-\hat{r})$ face. The magnitude of the Coulomb friction force is μ times the normal pressure $-\sigma_{rr}$ on that face; writing it this way presumes $\sigma_{rr} \leq 0$ (the element is under compression), which we verify self-consistently once the system is solved (Section 4). Therefore,

$$\sigma_{ur} \left(\zeta = -\frac{t}{2} \right) = -\mu \sigma_{rr} \left(\zeta = -\frac{t}{2} \right) \sin(2\phi), \quad (31)$$

$$\sigma_{vr} \left(\zeta = -\frac{t}{2} \right) = -\mu \sigma_{rr} \left(\zeta = -\frac{t}{2} \right) \cos(2\phi). \quad (32)$$

By the same argument, the friction exerted by the outer layer is the exact opposite, dragging the element forward, giving the corresponding relationship on the outer $(+\hat{r})$ face (the sign does not flip, due to the sign convention for shear stresses on opposite faces of an element):

$$\sigma_{ur} \left(\zeta = \frac{t}{2} \right) = -\mu \sigma_{rr} \left(\zeta = \frac{t}{2} \right) \sin(2\phi), \quad (33)$$

$$\sigma_{vr} \left(\zeta = \frac{t}{2} \right) = -\mu \sigma_{rr} \left(\zeta = \frac{t}{2} \right) \cos(2\phi). \quad (34)$$

Each pair holds only where the corresponding neighbor is actually present (Eq. 23–24); where the neighbor is absent, both shear components vanish along with the contact pressure that would otherwise drive them.

3.3 Layer-layer contact relationship

By Newton's third law, two surfaces in contact experience equal and opposite traction; because their outward normals point in opposite directions, this means their normal stresses must agree. It suffices to match σ_{rr} alone, since the transverse shears at each surface are already tied to it by Eqs. (31)–(34). This gives a recursive relationship for the boundary values of σ_{rr} , using the same

piecewise gates as Eq. (23)–(24):

$$\text{Outer surface: } \sigma_{rr} \left(u, v, \zeta = \frac{t}{2} \right) = \begin{cases} \sigma_{rr} (u^+, v^+, \zeta = -\frac{t}{2}) & v \geq -\frac{W}{2} + 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (35)$$

$$\text{Inner surface: } \sigma_{rr} \left(u, v, \zeta = -\frac{t}{2} \right) = \begin{cases} \sigma_{rr} (u^-, v^-, \zeta = \frac{t}{2}) & v \leq \frac{W}{2} - 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (36)$$

(the outer face of this element touches the *inner* face of its outer neighbor at (u^+, v^+) , and the inner face touches the *outer* face of its inner neighbor at (u^-, v^-) — note the two relations reference opposite neighbors, matching Section 3.1). Equations (35)–(36) recursively relate the boundary stress of one layer to that of its neighbor, one full turn away; propagated far enough in u , they reach the outermost wound layer at $u = L$, past which no material is wound at all (Section 3.1), so

$$\sigma_{rr} \left(u = L, v, +\frac{t}{2} \right) = 0 \quad \text{for all } v, \quad (37)$$

regardless of the v -condition in Eq. (35) — there is simply nothing at (u^+, v^+) to be in contact with. This is the base case that starts the recursion. Note that Eqs. (35)–(36) are pure *matching* conditions between neighbors: they say nothing yet about what value σ_{rr} actually takes away from this base case. Section 4 supplies the piece that does.

4 PDE formulation and closed-form solution

We now assemble the pieces above into a closed system for the stress field. The three equilibrium equations, Eqs. (15)–(17), are three PDEs for six unknown stress components at every point (u, v, ζ) . Section 3 supplies what is needed to close this at the two surfaces $\zeta = \pm \frac{t}{2}$: there, σ_{ur}, σ_{vr} are fixed directly in terms of σ_{rr} by Coulomb friction (Eqs. 31–34), and σ_{rr} ’s own two boundary values are tied recursively, layer to layer, by Newton’s third law (Eqs. 35–36). Combined with the linear-through-thickness approximation of Section 2, this reduces every unknown to a function of (u, v) alone. We also carry forward the free-edge boundary conditions on the long, unloaded edges of the ribbon,

$$\boxed{\sigma_{uv} \left(u, v = \pm \frac{W}{2} \right) = 0, \quad \sigma_{vv} \left(u, v = \pm \frac{W}{2} \right) = 0,} \quad (38)$$

which hold at every ζ and so apply directly to the thickness-uniform σ_{uv}, σ_{vv} used below, and the end load applied at the active end, $F(x) = W[\sigma_{uu}(L) \sin \phi + \sigma_{uv}(L) \cos \phi]$ (the resultant of σ_{uu}, σ_{uv} projected onto the global pulling direction).

4.1 Thickness integration

Integrate Eq. (15) over $\zeta \in [-\frac{t}{2}, \frac{t}{2}]$. Writing $\sigma_{ur}(u, v, \zeta)$ as linear in ζ with boundary values fixed by Eqs. (31),(33), its thickness average is

$$\bar{\sigma}_{ur} := \frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \sigma_{ur} d\zeta = \frac{1}{2} \left[\sigma_{ur}(\frac{t}{2}) + \sigma_{ur}(-\frac{t}{2}) \right]$$

(the linear part of the profile integrates to zero over the symmetric interval), while $\int \frac{\partial \sigma_{ur}}{\partial \zeta} d\zeta = \sigma_{ur}(\frac{t}{2}) - \sigma_{ur}(-\frac{t}{2}) =: f_u(u, v)$ is the same boundary-flux difference used previously. Taking σ_{uu}, σ_{uv} uniform through the thickness (nothing at this order sources a ζ -dependence in them) and dividing by t ,

$$\boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{f_u}{t} + \kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = 0,} \quad (39)$$

and, by the identical argument applied to Eq. (16),

$$\boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{f_v}{t} - \kappa_{uv} \bar{\sigma}_{ur} + \kappa_{vv} \bar{\sigma}_{vr} = 0.} \quad (40)$$

Unlike a treatment that discards the curvature-weighted terms outright, Eqs. (39)–(40) keep them, now expressed through the through-thickness *average* $\bar{\sigma}_{ur}, \bar{\sigma}_{vr}$ rather than the pointwise value. Using Eqs. (31)–(34), $\bar{\sigma}_{ur} = -\frac{\mu}{2} \sin(2\phi) [\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2})]$ and $\bar{\sigma}_{vr} = -\frac{\mu}{2} \cos(2\phi) [\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2})]$ (each term vanishing on whichever side has no neighbor), and a short calculation from Eq. (9) shows $\kappa_{uu} \sin 2\phi - \kappa_{uv} \cos 2\phi = \kappa_{uv}$, so the curvature-weighted correction in Eq. (39) collapses to

$$\kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = -\frac{\mu \kappa_{uv}}{2} \left[\sigma_{rr}(\frac{t}{2}) + \sigma_{rr}(-\frac{t}{2}) \right], \quad (41)$$

proportional to the *sum*, rather than the difference, of the two boundary pressures — and to κ_{uv} , not κ_{uu} . Because κ_{uv} vanishes at $\phi = 0, 90^\circ$ and is exactly the off-diagonal curvature responsible for load transfer between u and v (Section 1.3), this term is a genuine, if secondary, correction to the friction-driven balance — not something to be dropped outright — and we carry it forward symbolically, returning to its relative size once σ_{rr} ’s boundary values are pinned down below.

We now integrate the out-of-plane equation, Eq. (17), the same way. Its ζ -derivative term integrates directly to the *jump* in σ_{rr} across the thickness, $\Delta \sigma_{rr} := \sigma_{rr}(\frac{t}{2}) - \sigma_{rr}(-\frac{t}{2})$, while $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ (uniform through the thickness) simply pick up a factor of t :

$$\boxed{\Delta \sigma_{rr} = t [\kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} + \kappa_{vv} \sigma_{vv}] - t \left[\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v} \right],} \quad (42)$$

where the second bracket involves an extra in-plane derivative of an already boundary-traction-sized quantity and is subdominant to the first; we drop it below and return to it only if the leading balance proves insufficient. Equation (42) is the relation this section has been building toward: it says the two contact pressures felt on a ribbon element’s two faces cannot be equal in general, but must differ by an amount set directly by the local in-plane tension, weighted by curvature and by

the actual physical thickness t — exactly the capstan-type mechanism (normal pressure $\sim$ tension over radius, for a tensioned band pressed against a curved surface) that generates contact pressure from tension in the first place. Writing out $\Delta\sigma_{rr} = \sigma_{rr}(\frac{t}{2}) - \sigma_{rr}(-\frac{t}{2})$ explicitly, Eq. (42) becomes

$$\boxed{\sigma_{rr}(u, v, \frac{t}{2}) - \sigma_{rr}(u, v, -\frac{t}{2}) \approx t \left[\kappa_{uu}\sigma_{uu}(u, v) - 2\kappa_{uv}\sigma_{uv}(u, v) + \kappa_{vv}\sigma_{vv}(u, v) \right]}. \quad (43)$$

4.2 Closing the system: why σ_{rr} does not simply propagate to zero

Read on its own, the matching condition of Section 3.3 (Eqs. 35–36) contains no mechanism for generating a nonzero contact pressure anywhere: starting from the base case $\sigma_{rr}(L, v, \frac{t}{2}) = 0$, it would simply propagate that zero value backward through every layer, forcing $\sigma_{rr}(u, v, \pm\frac{t}{2}) \equiv 0$ everywhere, and hence, through Eqs. (31)–(34), $\sigma_{ur} \equiv \sigma_{vr} \equiv 0$ as well. It is Eq. (43) that resolves this: the matching condition and the pressure-jump relation hold *simultaneously*, not one after the other. Evaluating Eq. (43) at (u^+, v^+) and substituting $\sigma_{rr}(u^+, v^+, -\frac{t}{2}) = \sigma_{rr}(u, v, \frac{t}{2})$ (Eq. 35) gives

$$\boxed{\begin{aligned} \sigma_{rr}(u, v, \frac{t}{2}) &= \sigma_{rr}(u^+, v^+, \frac{t}{2}) - t \left[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv} \right] (u^+, v^+), \\ &\text{valid for } v \geq -\frac{W}{2} + 2\pi r \sin \phi, \end{aligned}} \quad (44)$$

with $\sigma_{rr}(L, v, \frac{t}{2}) = 0$ starting the recursion. Unlike the matching condition alone, Eq. (44) carries a genuine source term at every step, $-t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}]$, nonzero whenever the local in-plane tension is nonzero. Unwound from the base case, $\sigma_{rr}(u, v, \frac{t}{2})$ is a sum of such terms evaluated at every wound layer further out than (u, v) — so σ_{rr} accumulates precisely because, and only because, σ_{uu} (and σ_{uv}, σ_{vv}) are nonzero, exactly as anticipated. And σ_{uu} is nonzero in turn because the Instron pull at the active end prescribes a nonzero resultant force $F(x)$ there (Section 4, opening): it is that end condition, not anything internal to Eqs. (39)–(43), that ultimately keeps every stress component in the system from relaxing to zero. (One can check directly that Eq. (36) combined with Eq. (43) at (u, v) reproduces exactly Eq. (44) shifted by one layer, so the two contact relations of Section 3.3 are consistent with a single underlying recursion, not two independent ones.)

This also gives the complementary limit directly. If the helix is wound loosely enough that no v satisfies either overlap condition at the current x (Eq. 23–24), then $\sigma_{rr}(u, v, \pm\frac{t}{2}) \equiv 0$ is forced pointwise *by definition* rather than by Eq. (44), and Eq. (43) then *requires* $\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv} \equiv 0$ as a consistency condition. Together with $f_u = f_v = 0$ in Eqs. (39)–(40) and the free-edge conditions of Eq. (38), this drives $\sigma_{uu}, \sigma_{uv}, \sigma_{vv} \equiv 0$ as well: with no self-contact anywhere there is no mechanism to resist the pull, so the ribbon offers no static tension and simply slides freely. This matches the physical expectation exactly.

4.3 The accumulated contact pressure in closed form

Equation (44) steps $(u, v) \rightarrow (u^+, v^+) = (u + 2\pi r \cos \phi, v - 2\pi r \sin \phi)$ at each iteration — a fixed shift, since r, ϕ are frozen at any given extension x . Because the source term on the right of Eq. (44) carries an explicit factor of t , the change in $\sigma_{rr}(\cdot, \frac{t}{2})$ from one wound layer to the next is small (order $\frac{t}{r}$ relative to σ_{uu} itself), even though the geometric step $2\pi r \cos \phi$ between layers is not. This is

exactly the situation in which a discrete recursion is well approximated by its continuum limit: treating $\sigma_{rr}(u, v, \frac{t}{2})$ as a smooth function and expanding Eq. (44) to first order in the step gives the transport equation

$$2\pi r \cos \phi \frac{\partial \sigma_{rr}(u, v, \frac{t}{2})}{\partial u} - 2\pi r \sin \phi \frac{\partial \sigma_{rr}(u, v, \frac{t}{2})}{\partial v} = t \left[\kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} + \kappa_{vv} \sigma_{vv} \right]. \quad (45)$$

The left side is exactly $2\pi r \cos \phi$ times the derivative of $\sigma_{rr}(\cdot, \frac{t}{2})$ along the direction $(\cos \phi, -\sin \phi)$ — precisely the direction of the neighbor shift (u^+, v^+) itself, per Eq. (21)–(22). If, as adopted below (Section 4.4), $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ are taken independent of v , the right side of Eq. (45) depends only on u , and integrating Eq. (45) along its characteristic line from (u, v) out to the base case at $u' = L$ gives the closed form

$$\boxed{\sigma_{rr}(u, v, \frac{t}{2}) = -\frac{t}{2\pi r \cos \phi} \int_u^L \left[\kappa_{uu} \sigma_{uu}(u') - 2\kappa_{uv} \sigma_{uv}(u') + \kappa_{vv} \sigma_{vv}(u') \right] du'}, \quad (46)$$

with $\sigma_{rr}(u, v, -\frac{t}{2})$ recovered directly from Eq. (43). Notably, Eq. (46) is then independent of v as well, consistent with the v -independence assumed for $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ on the right. This closed form implicitly assumes the characteristic through (u, v) reaches $u' = L$ before its v -coordinate drifts outside $[-\frac{W}{2}, \frac{W}{2}]$ and exits through the side of the ribbon rather than its end; resolving that width-edge truncation exactly would replace the upper limit L with $\min\left(L, u + (v + \frac{W}{2}) \frac{\cos \phi}{\sin \phi}\right)$, a secondary correction we do not carry through below.

4.4 Narrow-ribbon reduction and the closed ODE

As in a beam or narrow-plate approximation, and consistent with $L \gg W$ already assumed throughout, we take $\sigma_{vv}(u, v) \approx 0$ and σ_{uu}, σ_{uv} approximately independent of v (uniform across the width) for the remainder of this section; unlike the compatibility-based argument used in an earlier pass on this document, this is now adopted directly as a modeling approximation for a long, narrow ribbon, not derived. Integrating Eq. (39) over $v \in [-\frac{W}{2}, \frac{W}{2}]$: the $\frac{\partial \sigma_{uu}}{\partial u}$ term gives $W \frac{d\sigma_{uu}}{du}$ (an ordinary derivative, since σ_{uu} is now a function of u alone); the $\frac{\partial \sigma_{uv}}{\partial v}$ term integrates, via the free-edge condition of Eq. (38), to $\sigma_{uv}(u, \frac{W}{2}) - \sigma_{uv}(u, -\frac{W}{2}) = 0$; and the $\frac{f_u}{t}$ term integrates using Eq. (31), (33) and the fact that $\sigma_{rr}(u, \cdot, \pm \frac{t}{2})$ is itself v -independent (Eq. 46), so only the presence/absence gates of Eq. (23)–(24) depend on v , each contributing a factor equal to the length of its band, $W - 2\pi r \sin \phi$:

$$\begin{aligned} \frac{1}{tW} \int_{-\frac{W}{2}}^{\frac{W}{2}} f_u dv &= -\frac{\mu \sin(2\phi) (W - 2\pi r \sin \phi)}{tW} \left[\sigma_{rr}\left(u, \frac{t}{2}\right) - \sigma_{rr}\left(u, -\frac{t}{2}\right) \right] \\ &= -\frac{\mu(W - 2\pi r \sin \phi)}{W} \sin(2\phi) \left[\kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} \right], \end{aligned} \quad (47)$$

using Eq. (43) (with $\sigma_{vv} \approx 0$) for the bracketed jump. The curvature-correction term of Eq. (41) instead needs the *sum* $\sigma_{rr}(u, \frac{t}{2}) + \sigma_{rr}(u, -\frac{t}{2})$, not the difference — and by Eq. (46) that sum is set by the pressure *accumulated over every layer out to* $u = L$, not by the local tension at u alone. Whether this nonlocal term is actually small compared to Eq. (47) therefore depends on how many

layers are wound outside u (it is not guaranteed small just because $t \ll r$); we drop it here to obtain a closed, local system, flagging it — honestly, not by assumption — as an open refinement, exactly as with the term already dropped in Eq. (42). Applying the identical steps to Eq. (40) gives the pair

$$\boxed{\begin{aligned}\frac{d\sigma_{uu}}{du} &= \frac{\mu(W - 2\pi r \sin \phi)}{W} \sin(2\phi) [\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}], \\ \frac{d\sigma_{uv}}{du} &= \frac{\mu(W - 2\pi r \sin \phi)}{W} \cos(2\phi) [\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}].\end{aligned}} \quad (48)$$

(For $2\pi r \sin \phi > W$ — full overlap, no bare band at all — the coefficient $W - 2\pi r \sin \phi$ should be read as 0, not negative; we work in the regime $2\pi r \sin \phi < W$ below.)

4.5 Rank-one structure and closed-form solution

The right side of both equations in Eq. (48) is the same scalar, $\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv}$, multiplying $\sin(2\phi)$ in the first equation and $\cos(2\phi)$ in the second — so the 2×2 coefficient matrix is rank one, with $(\sin 2\phi, \cos 2\phi)$ its only nonzero-eigenvalue eigenvector. Writing $\sigma_{uu}(u) = A(u) \sin(2\phi)$, $\sigma_{uv}(u) = A(u) \cos(2\phi)$ (the general solution also admits a null-space term along $(2 \sin \phi, \cos \phi)$, which we set to zero, as before, for the minimal, best-conditioned solution carrying a single free constant), the bracket becomes

$$\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} = A(u) [\kappa_{uu} \sin 2\phi - 2\kappa_{uv} \cos 2\phi] = A(u) \frac{2 \sin^3 \phi \cos \phi}{r}$$

(using $\kappa_{uu} = \frac{\cos^2 \phi}{r}$, $\kappa_{uv} = \frac{\sin \phi \cos \phi}{r}$ from Eq. 9 and standard double-angle identities), and both equations of Eq. (48) collapse to the single scalar equation

$$\boxed{\frac{dA}{du} = \frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} A(u).} \quad (49)$$

Since r, ϕ are frozen at fixed x , the coefficient on the right is a constant in u , and Eq. (49) integrates immediately to

$$\boxed{A(u) = A_0 \exp \left[\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} u \right]}, \quad (50)$$

with $A_0 := A(0)$ the single free constant carrying the physical scale of the whole solution (the holding tension at the fixed, base end), so that

$$\boxed{\begin{aligned}\sigma_{uu}(u) &= A_0 \sin(2\phi) \exp \left[\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} u \right], \\ \sigma_{uv}(u) &= A_0 \cos(2\phi) \exp \left[\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} u \right].\end{aligned}} \quad (51)$$

4.6 Sign and monotonicity of the contact pressure

Section 3.2 assumed $\sigma_{rr} \leq 0$ (compression) throughout, in order to write the Coulomb friction magnitude as μ times $-\sigma_{rr}$; we now verify this self-consistently. Take $A_0 > 0$ (a positive base tension). Since $\phi \in (0^\circ, 90^\circ)$, $\sin \phi, \cos \phi > 0$, so every prefactor appearing above is non-negative, and $\sigma_{uu}(u) = A_0 \sin(2\phi) e^{(\dots)u} \geq 0$ for all $u \in [0, L]$: the axial stress is tensile everywhere, as it must be for a ribbon that is only ever pulled, never pushed. Substituting Eq. (51) into the bracket of Eq. (46),

$$\begin{aligned} \kappa_{uu}\sigma_{uu}(u') - 2\kappa_{uv}\sigma_{uv}(u') &= A_0 \frac{2\sin^3 \phi \cos \phi}{r} \exp \left[\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} u' \right] \\ &\geq 0 \quad \text{for all } u' \in [0, L], \end{aligned} \quad (52)$$

so the integrand in Eq. (46) is non-negative throughout, its integral over $u' \in [u, L]$ is non-negative, and the leading minus sign in Eq. (46) gives

$$\sigma_{rr} \left(u, v, \frac{t}{2} \right) \leq 0 \quad \text{for all } (u, v), \quad (53)$$

confirming the compression assumed throughout Section 3. The same holds for $\sigma_{rr}(u, v, -\frac{t}{2})$, since

$$\sigma_{rr} \left(u, v, -\frac{t}{2} \right) = \sigma_{rr} \left(u, v, \frac{t}{2} \right) - t \left[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} \right] (u) \quad (54)$$

(Eq. 43) subtracts a term that is itself non-negative. Beyond the sign, Eq. (46) is manifestly *monotonic*: as u decreases from L (where it is exactly zero, by the base case) toward 0, the integration range $[u, L]$ only grows, so $|\sigma_{rr}(u, v, \frac{t}{2})|$ increases monotonically. This matches the physical picture directly: a layer near the base end ($u = 0$) has every other wound layer stacked outside it and so carries the full accumulated contact pressure, while a layer at the active end ($u = L$) has nothing wound outside it and carries none.

4.7 Force-displacement relation

The resultant force at the active end, projected onto the global pulling direction, is $F(x) = W[\sigma_{uu}(L) \sin \phi + \sigma_{uv}(L) \cos \phi]$ (Section 4, opening). Substituting Eq. (51) at $u = L$ and using $\sin(2\phi) \sin \phi + \cos(2\phi) \cos \phi = \cos(2\phi - \phi) = \cos \phi$ to collapse the double-angle dependence,

$$F(x) = W A_0 \cos \phi \exp \left[\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} L \right]. \quad (55)$$

Writing this out in terms of x via $\sin \phi = \frac{x}{L}$, $\cos \phi = \frac{\sqrt{L^2 - x^2}}{L}$, $r = \frac{\sqrt{L^2 - x^2}}{\theta_L}$ (Eq. 2–3), the exponent simplifies (the two factors of $\sqrt{L^2 - x^2}$, one from $\cos \phi$ and one from $\frac{1}{r}$, cancel) to

$$\frac{2\mu(W - 2\pi r \sin \phi) \sin^3 \phi \cos \phi}{rW} L = \mu \left(1 - \frac{2\pi x \sqrt{L^2 - x^2}}{\theta_L L W} \right) \frac{2\theta_L x^3}{L^3}, \quad (56)$$

giving the closed-form force–displacement law

$$F(x) = \frac{WA_0\sqrt{L^2 - x^2}}{L} \exp \left[\mu \left(1 - \frac{2\pi x\sqrt{L^2 - x^2}}{\theta_L LW} \right) \frac{2\theta_L x^3}{L^3} \right], \quad (57)$$

valid for $2\pi x\sqrt{L^2 - x^2} < \theta_L LW$ (Section 4.4), i.e., before the overlap band consumes the full width W . As in the leading-order picture of Section 4.2, $F(0) = \frac{WA_0}{L} \cdot L = WA_0$ is set entirely by the base tension A_0 , and the exponential growth in x is exactly the accumulated, capstan-type effect of Section 4.3: each additional bit of wound-over contact multiplies the required pulling force rather than adding to it.

5 Experimental validation

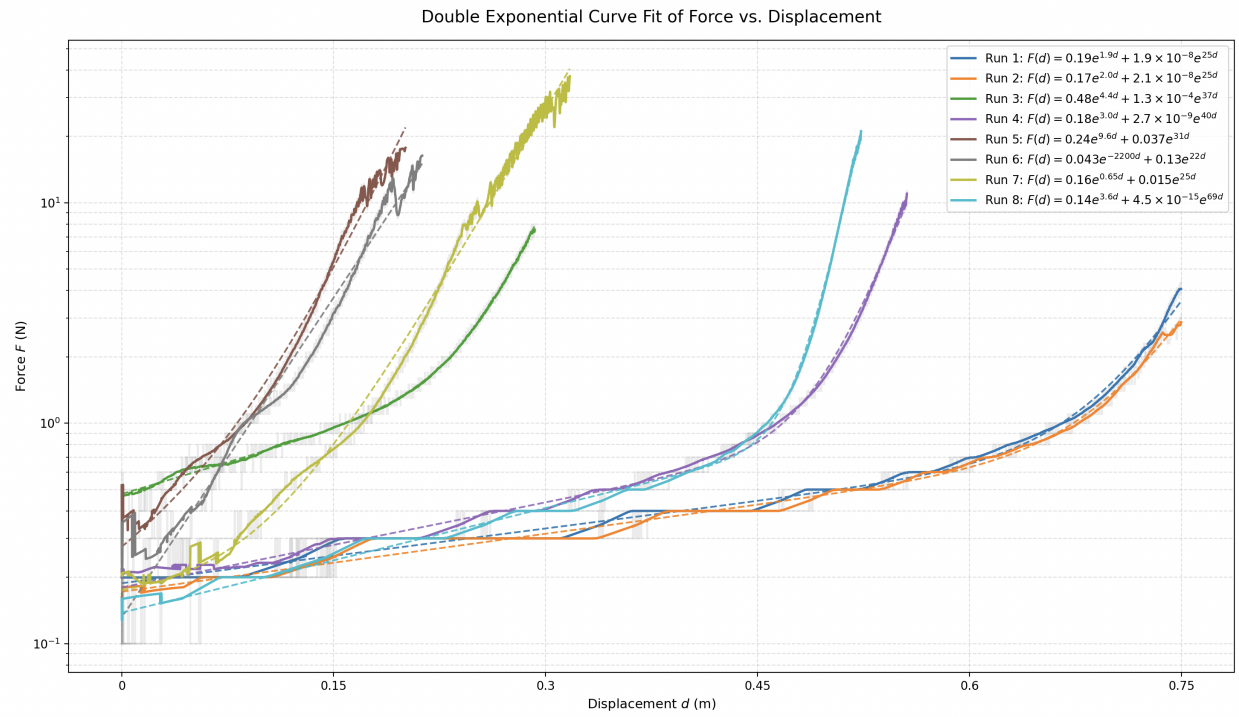


Figure 3: Instron force-displacement data, with raw quantized data (grey), smoothed data (solid colored), and curve fit (dashed colored).